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ADVPRINT PRO (ADVANCED PRINTER) · BY BIT DREAM IT — INNOVATIVE HEALTHCARE SOLUTIONS

v3.1.2 Build & Deploy Guide

Premium Graphical Window · THE ONE UI · Center Mode · step by step

Base	v3.0.9 (dropdown fix + per-printer IPP queues) → this release adds the UI, the window and Center Mode. Every previous feature ships unchanged.
THE ONE UI	The hub serves `console.html` at `http://127.0.0.1:9235/ui?token=…` — same page in the native window (gui_app v2), in Chrome and embedded in the extension options Console tab.
Center Mode	One center PC runs the tool; child PCs install ONLY the extension (or use plain IPP queues) and print through the center PC.
Build	`scripts/build-all.sh` builds CRX + extension zip + Setup EXE + DEB + SHA256SUMS from source alone (signing key in `extension-signing/`).
Tests	`python3 -m pytest tests/ -q` → 280 passed, 1 skipped.
Git	Tags `v3.0.9` and `v3.1.2` committed locally — push with `git push origin main --tags`.

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This kit: source zip + CRX + extension zip + Setup EXE + DEB + SHA256SUMS + release notes.

1 What you got (the release kit)

File	What it is	Use it on
`AdvPrint-Pro-Setup-3.1.2.exe` (≈10 MB)	Windows installer: hub + bundled Python runtime + CRX + START-CONSOLE window launcher + silent-extension policy + firewall rules.	Center PC (Windows)
`advprint-pro_3.1.2_all.deb`	Linux package: `/opt/advprint/hub/` (advprint.py, console.html, gui_app.py, CRX), systemd user service, autostart, policy.	Center PC (Linux)
`AdvPrint-Pro.crx` (signed)	CRX3 extension, id `iklldbkamojokkfdlcjibiljpgbbnohc`, v3.1.2 — served by the hub itself for silent install/auto-update.	Any Chrome/Edge/Brave
`AdvPrint-Pro-Extension-3.1.2.zip`	Unpacked extension for Web-store upload / Load-unpacked dev installs.	Dev / store
`SHA256SUMS-3.1.2.txt`	Checksums — verify with `sha256sum -c SHA256SUMS-3.1.2.txt`.	Everywhere
`AdvPrint-Pro-3.1.2-source.zip`	Full source tree (both hub copies identical, extension, installers, docs, tests, build scripts, signing key).	Build everything yourself

v3.0.9 fixes included: the Default-printer dropdown no longer shows doubled names ("HP Laser 103 107 108HP Laser 103 107 108") — one missing paren in `options.js` made the label evaluate to `name+name`; fixed + regression-tested. **v3.0.9 feature included:** every local printer gets its own queue `ipp://ip:9235/ipp/print/<printer>` + one mDNS service per printer.

2 Build everything from source — step by step

2.1 Requirements

Tool	Needed for	Install
python3 ≥ 3.9	hub, tests, CRX3 packer	preinstalled on Linux / python.org on Windows
zip	extension zip	`sudo apt-get install zip`
dpkg-deb	DEB package	preinstalled on Debian/Ubuntu
NSIS 3.x (makensis)	Setup EXE	`sudo apt-get install nsis`
Node.js (optional)	one dropdown regression test	the test skips cleanly without it

2.2 Run the tests first

```
cd AdvPrint-Pro
python3 -m pytest tests/ -q
# → 280 passed, 1 skipped
```

2.3 One command — all artifacts

```
bash scripts/build-all.sh
# dist/AdvPrint-Pro.crx
# dist/AdvPrint-Pro-Extension-<V>.zip
# dist/advprint-pro_<V>_all.deb
# dist/AdvPrint-Pro-Setup-<V>.exe       (needs makensis)
# dist/SHA256SUMS-<V>.txt
```

The script stages the hub files, the installers, the bundled Python runtime and the docs into `installer/stage/`, packs the CRX with the repo key, builds the DEB, runs makensis and writes checksums.

2.4 Setup EXE only (rebuild without rebuilding the rest)

```
cd installer && makensis -V2 setup.nsi
```

No sudo? NSIS local-prefix recipe (used for this very release):

```
apt-get download nsis nsis-common
dpkg -x nsis_*.deb ~/nsis-root && dpkg -x nsis-common_*.deb ~/nsis-root
cd installer && NSISDIR=$HOME/nsis-root/usr/share/nsis \
  LD_LIBRARY_PATH=$HOME/nsis-root/usr/lib/x86_64-linux-gnu \
  $HOME/nsis-root/usr/bin/makensis -V2 setup.nsi
```

2.5 CRX / extension only

```
python3 tool/crx3.py pack extension \
        extension-signing/advprint-printer.pem dist/AdvPrint-Pro.crx
python3 tool/crx3.py info dist/AdvPrint-Pro.crx     # verify id + version
cd extension && zip -r ../dist/AdvPrint-Pro-Extension.zip .
```

The key ships in the tree, so the extension id stays `likldbkamojokkfdlcjibiljpgbbnohc` — Chrome keeps auto-updating the force-installed extension from YOUR hub (`/extension/updates.xml`).

2.6 DEB only

```
sudo dpkg -i dist/advprint-pro_<V>_all.deb        # install
sudo dpkg -r advprint-pro                          # uninstall
```

3 Set up the CENTER PC (the one printing PC)

1. Install the tool: run `AdvPrint-Pro-Setup-3.1.2.exe` (Windows) or `sudo dpkg -i advprint-pro_3.1.2_all.deb` (Linux). The hub starts at `http://127.0.0.1:9235` (tray icon / systemd user service).
2. Open **THE ONE UI**: tray menu → **Console (THE ONE UI)**, or `START-CONSOLE.vbs`, or a browser on `http://127.0.0.1:9235/ui?token=<token>`. The native window opens the same page (pywebview → Qt → GTK → Chrome app window → browser ladder — no PySide6, installer ≈10 MB).
3. Console → **Settings** tab: turn **LAN sharing ON** (or tray → Allow LAN in firewall). Keep **Per-printer IPP queues** ON and **mDNS** ON. Save & Apply.
4. Console → **Extension** tab: copy the **token** — child PCs need it.
5. Verify: Console → **Printers** shows every printer with its own `ipp://…:9235/ipp/print/<printer>` queue; press **Test page**.

> **Firewall (Windows):** inbound **TCP 9235** (printing) + **UDP 5353** (mDNS discovery). The Setup EXE adds both; tray → "Allow LAN in firewall" re-adds TCP any time.

4 Set up the CHILD PCs (extension and/or IPP — no tool)

Way A — extension only (silent browser printing with rules)

1. Install the extension on the child PC: load `AdvPrint-Pro.crx` / the zip (chrome://extensions → Developer mode → Load unpacked), or let the center PC's force-install policy install it silently (domain PCs).
2. Extension → Options → **Connections** → **Center PC / child PC mode**:
   - **Mode**: `Auto (recommended)` — uses a local tool when present, otherwise the Center PC. Choose `Center PC only` for a pure child PC.
   - **Center PC hub URL**: `http://CENTER-IP:9235`
   - **Center PC token**: the token from step 4 above → **Save settings**.
3. The status line proves it: `[auto] local tool NOT running → printing on CENTER PC: http://…`. Popup → **Console** opens the center's THE ONE UI.

Way B — per-printer IPP queues (no extension at all)

1. **Windows 10/11**: Settings → Printers → Add device — the center's printers appear by themselves (mDNS). Or Add manually → "Select a shared printer by name" → `ipp://CENTER-IP:9235/ipp/print/<printer>`.

2. **Linux/CUPS**: `lpadmin -p advprint-hp -E -v ipp://CENTER-IP:9235/ipp/print/HP-Laser-103-107-108 -m everywhere`
3. **macOS / iPhone / Android**: discovered automatically (Bonjour / AirPrint / Mopria).
4. Queue names accept the raw Windows name (URL-encoded) or the slug form — both work. The Console → **IPP Printer** tab lists every URL with a Copy button.

> **Both ways combined is fine:** extension for silent browser printing + rules, IPP queues so ANY application on the child PC prints on the center PC too. Many child PCs can connect to one center PC at the same time. Full details: `docs/CENTER-CHILD-PCS.md`.

5 Verify everything (10-minute checklist)

#	Check	Expected
1	Console: `http://127.0.0.1:9235/ui?token=…`	IDM console, 8 tabs live, status bar "hub running · v3.1.2"
2	Console → Test page (Dashboard)	paper comes out; job shows PRINTING → PRINTED
3	Extension Default-printer dropdown	**every name exactly ONCE** (v3.0.9 fix) with [status] / (OS default)
4	Console → IPP Printer tab	every printer has `ipp://…/ipp/print/<printer>`
5	Child PC print (extension)	job lands in the CENTER PC's queue + history
6	Child PC IPP queue	appears in Windows printer list; prints on the center
7	Native window: START-CONSOLE.vbs / tray	pywebview window opens /ui (or falls down the ladder)
8	`sha256sum -c SHA256SUMS-3.1.2.txt`	all OK

6 Publish to GitHub (release process)

1. The tags are already committed locally: `git tag -l` → v3.0.9, v3.1.2. Push:

```
git push origin main --tags
```

2. GitHub → Releases → **Draft a new release** → choose tag `v3.1.2`.
3. Title: **PREMIUM GRAPHICAL WINDOW + CENTER MODE**. Description: paste `docs/RELEASE-NOTES-v3.1.2.md` (v3.0.9: paste `docs/RELEASE-NOTES-v3.0.9.md`).
4. Attach the 5 files from `dist/` (CRX, zip, EXE, DEB, SHA256SUMS) + the source zip → **Publish**.

5. Chrome auto-updates the force-installed extension on every child PC at next browser start (the hub serves `/extension/updates.xml`).

Bumping the version next time (touch every anchor together)

- `tool/unix/hub/advprint.py` AND `tool/windows/portable/hub/advprint.py`: `VERSION`, `VERSION_NAME` (keep both copies byte-identical — copy unix → windows).
- `extension/manifest.json`, `popup.html` (ver chip), `content/api.js` (fallback), `options.html` (verchip).
- `VERSION.txt`, `installer/setup.nsi` (OutFile, VIProductVersion, DisplayVersion), `installer/deb/DEBIAN/control`.
- `tests/test_hub_v2.py` version anchors + the release notes doc `docs/RELEASE-NOTES-v<V>.md`.

7 Troubleshooting

Symptom	Cause → Fix
Child PC cannot reach the hub	LAN sharing OFF → Console → Settings → LAN sharing ON, restart; firewall → tray "Allow LAN in firewall"; wrong token → copy from Console → Extension.
Printer not discovered on phones	mDNS OFF or firewall UDP 5353 → Settings → mDNS ON; the center must be ON the same LAN (no VPN).
IPP queue unknown (404)	Typo in the queue name — copy the exact URL from Console → IPP Printer. Names are case/punctuation-insensitive.
Console shows offline card	Hub not running → start AdvPrint Pro (tray / START-CONSOLE.vbs); extension Console tab → Retry now.
Native window falls back to browser	No GUI backend found — `pip install pywebview` for the WebView2/WebKit window (the ladder also uses an existing Chrome/Edge automatically).
makensis not found	Use the local-prefix recipe in §2.4, or `sudo apt-get install nsis`.
Extension still shows old version	Force-installed from the hub → restart Chrome; manual install → chrome://extensions → Update.

Where everything lives in the source tree: hub = `tool/unix/hub/advprint.py` + `console.html` + `gui_app.py` (identical copies under `tool/windows/portable/hub/`) · extension = `extension/` · installer = `installer/setup.nsi` + `installer/deb/` · build = `scripts/build-all.sh` · tests = `tests/` (280 passed) · docs = `docs/` (BUILD-FROM-SOURCE, CENTER-CHILD-PCS, IPP-GUIDE, MULTI-PRINTER).